Task 2: 10-Word Speech Recognition under Noise

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Abstract

This report studies how noise affects a simple 10-word speech recogniser. A small CNN on log-mel spectrograms, trained only on clean speech from Google Speech Commands, reaches 95.7 % accuracy on unseen speakers. The test recordings were mixed with white Gaussian and real background noise at 30 to −10 dB SNR. Accuracy stays above 86 % down to 10 dB and nears chance at −10 dB; white noise is more damaging than real noise at equal SNR, and noise biases the model toward *off* and *go*.

1. Dataset preparation

The data is a subset of Google Speech Commands v0.02 [1]: one-second single-word recordings by volunteers. Words: *yes, no, up, down, left, right, on, off, stop, go*; mono 16-bit PCM at **16 kHz**; **38,546 recordings from 2,492 speakers**, 3,723–4,044 per word (balanced within 5 % of the mean).

Speaker-independent split. The speaker ID is the file-name prefix before `_nohash_`. Speakers were shuffled with a fixed seed and assigned whole: 20 % to test (498), 10 % to validation, used only to pick the best epoch (249), and 70 % to training (1,745). A check in the code confirms that no speaker is in two sets and every word is in every set.

Set	yes	no	up	down	left	right	on	off	stop	go	Total
Train	2852	2838	2617	2741	2710	2667	2718	2642	2717	2733	27,235
Validation	388	366	364	389	351	378	337	351	377	369	3,670
Test	804	737	742	787	740	733	790	752	778	778	7,641
Total	4044	3941	3723	3917	3801	3778	3845	3745	3872	3880	38,546

Table 1. Recordings per word and set.

2. Feature extraction

Each recording is represented as a **log-mel spectrogram**: energy over time in bands spaced like human hearing, on a log scale that evens out loudness. It shows vowel formants, fricative noise and plosive bursts as a small image that suits a CNN.

Preprocessing and parameters. No resampling was needed (all clips are 16 kHz). Every waveform was set to exactly 1 s (16,000 samples) by zero-padding shorter clips equally on both sides. Settings, identical for all recordings and conditions: **25 ms frames (400 samples), 10 ms hop (160 samples), Hann window, 512-point FFT, 40 mel filters from 20 Hz to 8 kHz**, power spectrum, then the natural logarithm ($+10^{-6}$). With centred frames this gives a **40 × 101** matrix (mel bands × frames) per recording. Each mel band is then standardised with the mean and standard deviation of the training set.

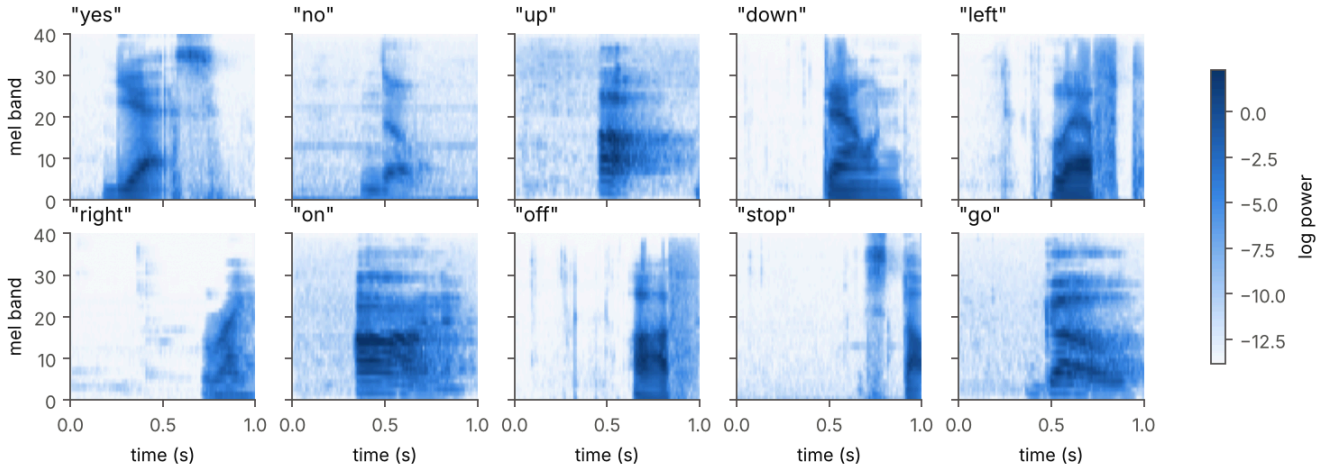


Figure 1. Log-mel spectrograms, one training example per word (darker = more energy).

Interpretation ("yes"). The word starts around 0.2 s with the /jɛ/ vowel: its gliding formants form slanted bands in mel bands 0–10 and 20–35. From about 0.6 s the energy moves to the top bands (30–40): the /s/ fricative, with almost no low-frequency energy. Other words show their own patterns, such as the separate /s/ and vowel bursts of "stop".

3. Model

Architecture (98,682 trainable parameters). One small CNN, treating the spectrogram as a one-channel image, was used for all experiments: four convolutional blocks, each a 3×3 convolution, batch normalisation, ReLU activation and 2×2 max-pooling, with 16, 32, 64 and 128 filters. The feature map shrinks from 40 × 101 to 128 × 2 × 6. Global average pooling then gives a 128-value vector, followed by dropout (0.3) and a fully connected output layer with 10 units and softmax.

Training. Cross-entropy loss, Adam optimizer (learning rate 0.001, weight decay 10^{-4}, cosine learning-rate decay), 20 epochs, batch size 64. No noise or other augmentation was applied to the training data. The weights from the epoch with the best validation accuracy were kept (epoch 18: 97.0 % validation, 99.2 % training accuracy). Train/test: 27,235/7,641 recordings from 1,745/498 disjoint speakers.

4. Testing under noise

Procedure. Noise was added only to the test waveforms, before feature extraction; the trained model was never changed. SNR = $10 \cdot \log_{10}(P_{speech} / P_{noise})$, with P_{speech} the mean power of the original recording; the scaled noise covers the whole clip. The same 7,641 test recordings were used in three conditions: (1) **clean** speech; (2) **white Gaussian noise**; (3) **real background noise** from the four recordings supplied with the dataset (*doing the dishes, cat miaowing, exercise bike, running tap*). In the main "real (mixed)" condition each clip gets one of the four at random, and each recording was also tested alone. SNRs: 30, 20, 10, 0, −5, −10 dB, with the same noise sample per clip at every level; the negative levels were added because SNR is averaged over the whole clip, pauses included.

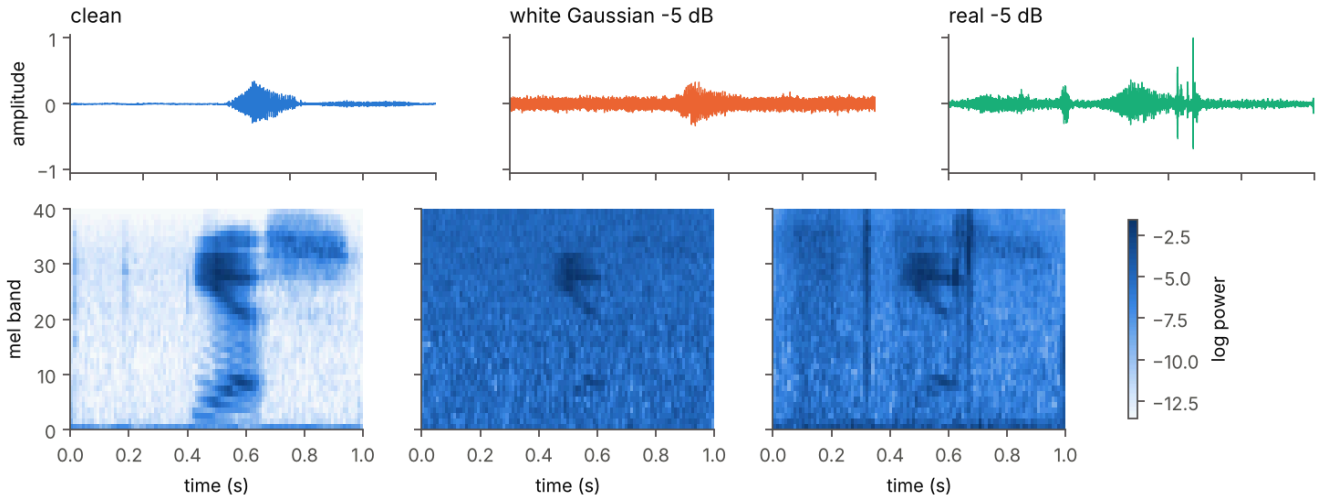


Figure 2. A test recording of "yes": clean, with white noise and with real noise (cat miaowing) at −5 dB. White noise masks all but the strongest vowel formants and removes the final /s/; the bursty real noise leaves more of the word visible.

Noise	30 dB	20 dB	10 dB	0 dB	−5 dB	−10 dB
White Gaussian	94.9	92.1	86.3	63.6	36.2	17.6
Real (mixed)	95.2	92.6	87.7	69.2	45.9	24.0
Doing the dishes	95.5	94.3	89.0	64.4	36.7	18.0
Exercise bike	95.2	92.2	87.7	65.4	37.4	17.8
Running tap	95.0	92.7	88.7	75.0	55.8	28.1
Cat miaowing	94.7	91.9	86.6	73.3	57.0	33.7

Table 2. Test accuracy (%) by noise type and SNR. Clean accuracy: 95.7 %; chance: 10 %.

SNR. Clean accuracy is **95.7 %**. Down to 10 dB the model loses less than 10 percentage points; between 10 and −10 dB accuracy falls steeply, to 17.6 % with white noise, close to chance.

Noise type. At high SNR all noise types are equivalent. From 0 dB down, white noise is the most damaging: mixed real noise keeps 5.6 points more at 0 dB and 9.7 points more at −5 dB. The dishes and the exercise bike are almost as harmful as white noise, while the running tap and the cat leave 75.0 % and 73.3 % at 0 dB. Bursty noise leaves clean gaps; white noise covers every frequency at every moment.

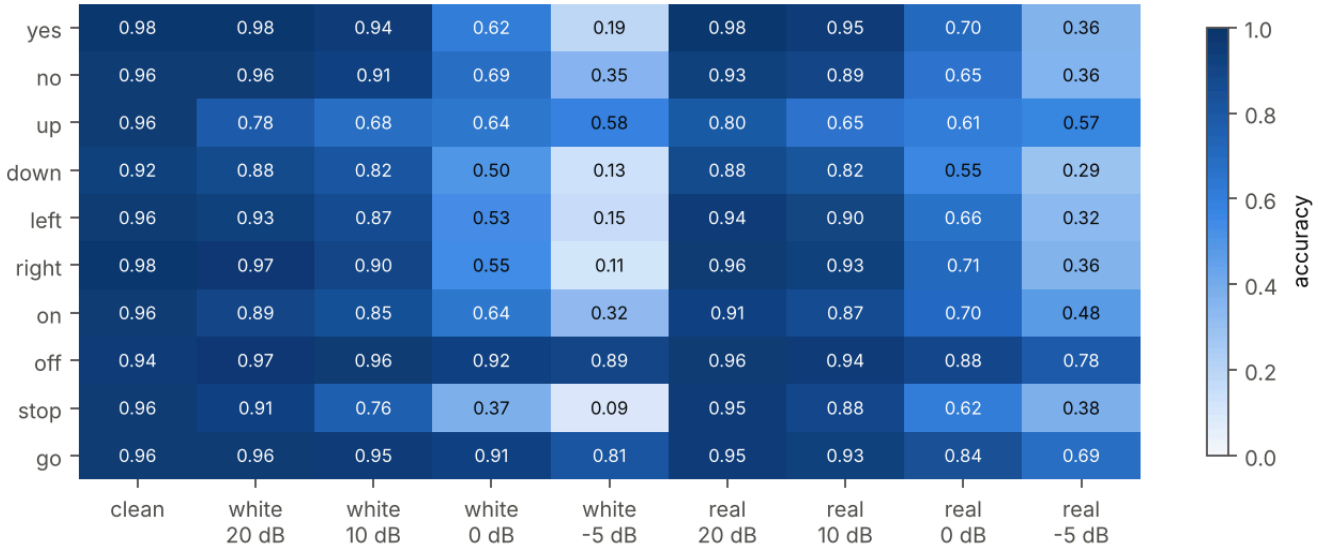


Figure 3. Accuracy of each word (recall) on clean speech and under white and mixed real noise.

Word identity. On clean speech every word scores 92–98 %. Under noise *stop, down, left* and *right* fall fastest (Figure 3): at 0 dB white noise *stop* drops to 37 %, as its /s/, /t/ and /p/ are weak, noise-like sounds. *Off* and *go* look robust (92 % and 91 %), but this is a bias: at 0 dB white noise the model outputs *off* 1,720 times and *go* 1,855 times, though each word has under 800 clips. At −10 dB, 60 % of all clips are labelled *off*.

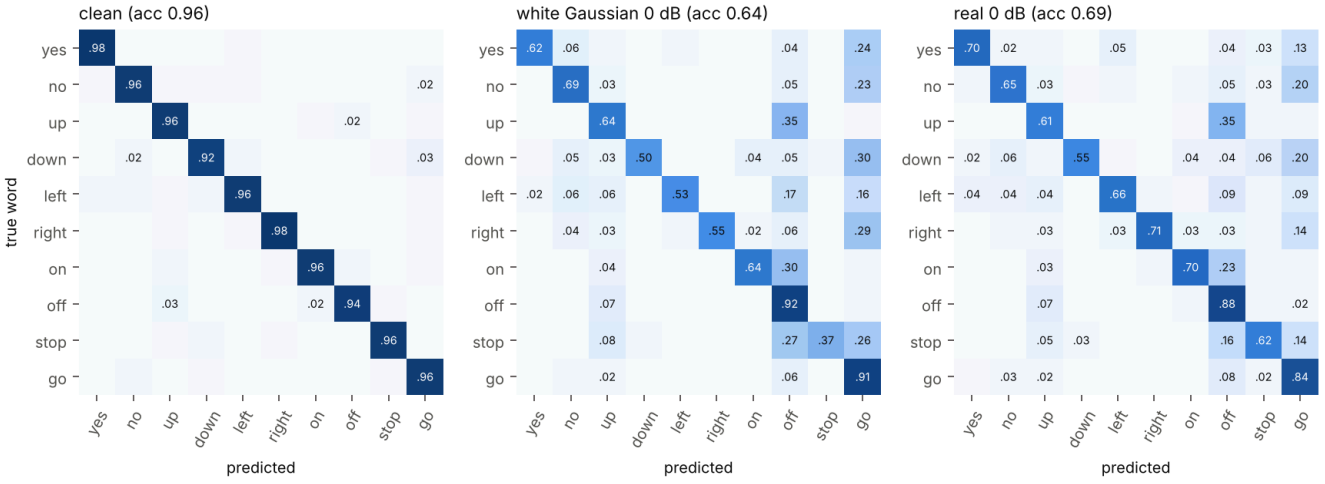


Figure 4. Row-normalised confusion matrices: clean speech, white noise and real noise at 0 dB.

Confusions. Clean-speech errors are rare and occur between similar-sounding words: *down → go* (26 cases), *off → up* (21), *down → no* (19), *off → on* (16), *up → off* (16). These pairs share a vowel or differ in a quiet final consonant. Noise amplifies them and pulls errors toward the bias words: *up → off* grows from 16 cases to 260 at 0 dB white noise, *on → off* from 10 to 239, followed by *down → go* and *right → go*. With real noise, errors at −10 dB spread over *off, go* and *up* instead of mainly *off*.

5. Conclusions

1. **SNR:** a simple CNN on log-mel features reaches 95.7 % on clean speech from unseen speakers and loses under 10 percentage points down to 10 dB. Below 0 dB accuracy collapses, reaching near chance at −10 dB.
2. **Noise type:** at equal SNR, stationary broadband white noise is the most damaging. Bursty or narrower real noise (cat, running tap) leaves up to 21 points more accuracy at −5 dB, while the dishes and the exercise bike are almost as harmful as white noise.
3. **Words:** words with weak consonants (*stop, down, left, right*) fail first. Noise biases the model toward *off* and *go*, so their apparent robustness is an artefact, and the most frequent noisy confusions (*up → off, on → off, down → go*) are clean-speech confusions amplified.
4. **Outlook:** the model saw only clean speech in training. Adding noise to the training data would likely improve robustness, but this was outside the scope of the task, which kept the model fixed and varied only the test conditions.

References

1. P. Warden, "Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition," arXiv:1804.03209, 2018.